

SIEGE // STUDY TERMINAL • CLASSIFIED OPERATIONS

THE OPERATOR'S FIELD MANUAL

How to run the terminal at one hundred percent — and take the highest marks on the planet, on purpose, by system rather than by luck.

OPERATOR: _____ | ISSUE 1.0 | CAIE / EDEXCEL IGCSE & A-LEVEL
DOCTRINE: ACTIVE RECALL • SPACED REPETITION • INTERLEAVING • PAST-PAPER SUPREMACY • MARK-
SCHEME FLUENCY

00 The Doctrine: Marks Are a System

The student who panics studies harder. The operator studies the exam.

Top marks are not a personality trait or a gift. They are the output of a **closed feedback loop**: do real questions → see exactly where you bleed marks → repair that precise gap → prove the repair on a fresh paper → repeat until the bleeding stops. SIEGE exists to run that loop faster than you could by hand, and to never let you lie to yourself about where you actually stand.

The five laws everything in this manual obeys

LAW	WHY IT WINS MARKS
1. Active recall > re-reading	Pulling an answer out of your head (not recognising it on a page) is what the exam tests and what builds durable memory. Every SIEGE mode is built to make you <i>produce</i> , not passively review.
2. Spaced repetition	Memory decays on a curve. Reviewing just before you'd forget resets the curve higher for less effort. The flashcard boxes and the Decay Curve chart are this law made visible.
3. Past papers are the syllabus	Examiners reuse question shapes, traps, and mark-scheme wording. Drilling <i>real</i> papers teaches the actual game. SIEGE only ever indexes real CAIE/Edexcel codes — it never invents questions.
4. The mark scheme is the rulebook	Marks are awarded for specific words and structures, not vibes. You must learn to think like the marker. MARK SCHEME and COMMAND WORDS modes drill exactly this.
5. Measure or it didn't happen	What you don't log, you can't fix. Logging every score and every wrong question turns "I think I'm okay at this" into a number you can attack.

THE ONE-SENTENCE STRATEGY

Convert every weakness into a logged number, let the Analytics terminal rank those numbers by yield, and spend tomorrow on whatever sits at the top of that list — every day, until the exam.

The rest of this manual is just the machinery for executing those five laws without friction. Read it once end-to-end, then keep it on your iPad and re-read Page 10 every Sunday.

01 Zero to Armed: First-Boot Setup

Ten minutes of setup decides whether the next ten weeks compound or leak.

Step 1 — Run the onboarding wizard honestly

When you first launch SIEGE it walks you through your **operator profile**. The single most important screen is **subject strengths**: rate each subject by how you *actually* perform, not how you wish you did. The whole engine — missions, the mastery radar, the task screener — weights your time toward what you rate weak. Lie here and you'll study what's comfortable instead of what's failing.

Step 2 — Set the clock

- **Exam session & date** — e.g. `0/N 2026` . This drives the **Days-Left** counter and Burn-Risk maths.
- **Deadline** — your internal "ready by" date, ideally 2–3 weeks before the real paper so you have a repair buffer.
- **Daily hours** — be realistic; the mission generator sizes each day's plan to this budget.

Step 3 — Connect the brain (Groq API key)

SIEGE's AI runs on Groq, which has a **free tier with no credit card**. Go to `console.groq.com/keys` , create a key, paste it into **Settings** → **groq api**, and hit **save & test** until it says *groq connected*.

MODEL	USE IT FOR
<code>gpt-oss-120b</code> (default)	Strongest reasoning — your everyday driver for explanations, mark schemes, planning.
<code>llama-3.1-8b-instant</code>	Fast, cheap, for quick lookups and high-volume flashcard generation.
<code>llama-4-scout</code> (auto)	Vision. The moment you attach an <i>image</i> of a question, SIEGE auto-routes to this so OCR Solver works.

BUDGET REALITY

On the free tier you can study for months without paying. If you ever hit the rate limit, add a card for the Developer tier — **\$0 minimum**, 10× the limits, and a 25% discount — not a lump-sum prepay.

Step 4 — Prove it works

Open the AI panel in **PLAN** mode and type *"build me today's mission."* If a real plan comes back referencing real paper codes, the terminal is armed. You're ready.

02 The Daily Battle Rhythm

Champions don't have more willpower. They have a loop they don't have to decide on.

Open SIEGE the same way every day and follow the same five-beat loop. The point is to remove decisions — you should never sit down wondering "what do I study?" The terminal already knows.

BEAT	WHERE	WHAT YOU DO
1 • Brief	Dashboard / Mission	Read today's auto-generated mission. It already weights weak subjects and names specific real papers.
2 • Strike	Pomodoro	Run focus blocks. Each completed block logs your minutes and feeds the streak — this is how the analytics get fuel.
3 • Drill	Past Papers	Do a real paper (or a timed section). Use Timer Overlay + Blind Mode for exam realism.
4 • Log	Paper modal	Mark it, enter your score, and — non-negotiable — list every wrong question number . This is the gold.
5 • Lock in	Flashcards / Mistakes	Turn each mistake into a flashcard or a mistake-log entry so it returns at the right time.

A model study day (4-hour budget)

BLOCK	ACTIVITY	MODE / TOOL
0:00–0:10	Brief + warm-up recall	Mission + due flashcards
0:10–1:00	Weak-topic teaching	EXPLAIN / CONCEPT TUTOR / SOCRATIC
1:00–2:15	Timed past-paper section	Past Papers + Timer Overlay
2:15–2:45	Mark & interrogate errors	MARK SCHEME + ROAST ME
2:45–3:30	Repair the exact gaps	Topic search → targeted papers
3:30–4:00	Encode + review telemetry	Flashcards + Analytics

THE STREAK IS NOT DECORATION

The streak/heatmap on the Analytics page is a commitment device. Protect it. A short logged session beats a heroic un-logged one, because only logged time becomes data the engine can act on.

03 Weaponising the AI: All 15 Modes

A generic chatbot answers questions. A mode-switched tutor runs a method.

Each tile on the AI screen loads a different specialist persona with its own system prompt. Pick the mode that matches the *job*, not just the subject.

MODE	FIRE IT WHEN...
PLAN	You need a study plan or want to add/remove tasks from today's mission by chat.
EXPLAIN	A concept won't click — paste it, get a clean breakdown.
MARK SCHEME	You've written an answer and want it graded like CAIE would, with mark-by-mark feedback.
WEAKNESS AUDIT	You want it to find what to fix next from your patterns.
RECOMMENDER	You're unsure which paper to attempt — get a targeted pick.
CONCEPT TUTOR	You want a full guided walk-through of one syllabus topic.
COMMAND WORDS	You lose marks on "explain vs describe vs evaluate." Drill the exact verbs.
SOCRATIC	You learn best by being asked, not told — it questions you to the answer.
CHEM 0620	Chemistry specialist: equations, moles, electrolysis, examiner traps.
TOPIC PREDICTOR	You want a pattern-based read on what's likely to appear this session.
OCR SOLVER	Upload a <i>photo</i> of a question — it reads and solves it (auto-uses the vision model).
ROAST ME	You want your answer torn apart so the real exam can't.
KEYWORD EXTRACT	Pull the mandatory mark-scheme keywords from a topic or answer.
DICTATE	Speak your answer instead of typing it.
EXAMINER DEBATE	Argue for partial marks — train yourself to justify every point.

POWER COMBO

MARK SCHEME → **ROAST ME** → **KEYWORD EXTRACT** on the same answer: get the honest grade, get it brutally critiqued, then extract the exact words you missed and turn them into flashcards. That three-punch is how a 6/10 answer becomes a 9/10 by next week.

04 Past-Paper Warfare

The single highest-yield activity in existence. Everything else serves this.

The Past Papers section is a grid of **real, indexed** CAIE/Edexcel codes. Every card carries direct **qp** / **ms** / **sme** links. Click a card to open the **paper intel** panel.

The paper-intel modal — your pre-flight check

- **Intel readout** — decoded component, marks, time, the real sitting (e.g. "Oct/Nov 2020"), variant/zone, and your status.
- **Open paper** — Question Paper / Mark Scheme links that go straight to the real PDF.
- **Log it** — set status, enter score/total, and list **wrong question numbers** + reflection notes. This is what powers your analytics.

✦ Ask AI for a topic

Pick the subject first, then hit **+ ask ai for topic** and name a precise topic ("electrolysis", "sound waves"). SIEGE asks an examiner-model which real papers most likely contain it, validates every code against the actual grid, and **filters the grid down to those papers** — each flagged with a likely question and reason. Treat results as a strong lead, not gospel: it's an examiner's best estimate, but every paper it shows is real and one click from the PDF.

The tactical toolbar

TOOL	WHAT IT DOES
Blind Mode	Hides answers/marks so you can't cheat yourself during a drill.
Timer Overlay	Real exam clock on screen — build time discipline.
MS Highlighter	Surfaces the mark-scheme keywords to internalise.
Curve Estimator	Projects your raw score onto grade boundaries.
Print Exam Booklet	Generate a clean printable set for offline drilling.
AI Forensics Pipeline	Drop a solved PDF: Auto-Grade, Predict Topics, Cognitive Replay, Syllabus Gaps, Examiner Report.

TRANSFER TO IPAD

Because paper links are public PDFs, you never need the laptop's network. Copy the **qp** link into a self-chat (Telegram Saved Messages) or scan a QR — open on the iPad and import into your notes app. Batch a whole session's links in one message.

05 The Analytics Terminal: Read Your Own Telemetry

Feelings lie. The dashboard doesn't. Every red number is a marked target.

The Analytics terminal computes everything live from your real logs. It shows **N/A** instead of faking numbers when there's no data — so the cure for an empty dashboard is always the same: **log more papers and study sessions.**

METRIC	WHAT IT MEANS → WHAT TO DO
A* Probability monte carlo	Simulates 1,500 exam outcomes from the spread of your real paper scores vs the A* boundary. Low? Your <i>average</i> or your <i>consistency</i> is short — drill until both rise.
Subject Mastery Radar	Blended score (topic progress + recent paper average) per subject. Attack the shortest spoke first; that's where marks are cheapest.
Task Screener	Your oracle. Ranks every weak topic by yield = (1 – mastery) boosted by logged mistakes . The top row is literally what to study next.
Effort vs Yield	Hours spent vs mastery per subject. A subject low-right = you grind but don't gain — change <i>method</i> , not hours.
SRS Retention & Decay	Predicted recall from your flashcard boxes, projected 30 days out. Falling? You're adding cards faster than you review them.
Global Velocity	Study hours/day vs last week. The momentum gauge — keep the arrow green.
Streak / Commit Heatmap	Consistency at a glance. Gaps are where progress evaporated.

THE 60-SECOND DAILY RITUAL

Open Analytics once a day. Read **one** number off the Task Screener's top row, and make that your first repair block tomorrow. That single habit — letting the data pick your target — is the difference between busy and effective.

The scrolling ticker up top is your situational awareness: overall mastery, retention, volatility (consistency of scores), burn-risk, days left. Glance at burn-risk weekly — if it climbs, your coverage is losing the race against the clock.

06 Memory & Mistakes: The Compounding Engine

A mistake you log once is worth more than a topic you re-read five times.

Flashcards — the Leitner ladder

Cards live in boxes 1–5. Grade honestly after each: **easy** jumps the card two boxes, **good** one, **again** resets it to box 1. Higher box = longer until you see it again. The discipline is in grading truthfully — a card you "sort of" knew is a *good*, not an *easy*.

- **Leech Hunter** — auto-suspends cards stuck in box 1. A leech means the card is badly written: break it into smaller atomic facts.
- **Cram Mode** — disables spacing and loads the whole deck. Reserve this for the final 48 hours only; it's a last resort, not a method.
- **Build cards from your errors**, not from the textbook. Every wrong exam question → one card.

The Mistake Log — your personal weakness map

Log every question you got wrong with its topic. Over weeks this becomes a heat-map of exactly where you bleed marks — and it feeds the Task Screener's yield ranking. Use **Redemption Mode** to re-attempt your past mistakes; a mistake isn't fixed until you've beaten a similar question cold.

The Notes Vault

Markdown notes per topic, supercharged by AI: **Auto-Summarize** condenses, **Generate Mind Map** restructures, **Cloze Test** turns notes into fill-the-blank active recall, and **AI Content Audit** cross-checks your notes against the syllabus to flag *missing* concepts — the gaps you don't know you have.

THE COMPOUNDING LAW

Re-reading is linear; spaced recall is exponential. Twenty minutes of due-card review every day will, by exam week, have moved more marks than any single all-nighter ever could. Trust the boxes.

Do

- Grade flashcards honestly
- Log 100% of wrong questions
- Review due cards *before* new study
- Convert every error into a card

Don't

- Mark "almost" as easy
- Skip logging a "silly" mistake
- Let due cards pile up
- Cram before the final 48h

07 The 12-Week Campaign & Exam-Week Protocol

Coverage early, repair in the middle, simulation at the end. In that order.

PHASE	WEEKS	MISSION
Foundation	1–3	Teach every weak topic to a baseline. Lean on CONCEPT TUTOR, EXPLAIN, SOCRATIC. Build your first flashcard decks.
Coverage	4–6	One real past paper per subject per week. Log everything. Let the radar fill in and the Task Screener wake up.
Saturation	7–9	Past-paper volume. Timed, Blind Mode on. Use Topic Search to hammer your two weakest radar spokes.
Repair	10–11	Pure Task-Screener-driven. Each day = the top weakness only, then a paper to prove the fix.
Simulation	12	Full papers under real exam timing. Curve Estimator after each. Only light, targeted review.

When it's going wrong — Crisis Mode

If the deadline is close and coverage is thin, Burn-Risk will spike. Trigger **Crisis Mode** and obey the Task Screener ruthlessly: maximum-yield topics only, drop everything already green, accept that "good enough across the board" beats "perfect in one corner."

Exam-week protocol

WHEN	PROTOCOL
T-7 days	Stop learning new topics. Switch fully to past papers + due-card review.
T-2 days	Re-attempt your logged mistakes (Redemption Mode). Run a COMMAND WORDS drill.
Night before	Light review of box-1/box-2 cards only. Pack equipment. Sleep — it's a mark-multiplier.
Morning of	No new material. 10 minutes of confident, already-known recall to warm up. Walk in calm.

IN THE EXAM HALL

Read the command word first, every time. "Describe" ≠ "Explain" ≠ "Evaluate." Answer the verb that's actually written, hit the mark-scheme keywords you drilled, and watch the clock you trained with.

08 The Operator's Twelve Commandments

Pin this page. Re-read it every Sunday. It is the whole manual compressed into twelve lines.

1. **Drill real papers above all else.** Nothing else moves marks as fast.
2. **Log every score and every wrong question.** Unlogged work is invisible to the engine and, soon, to you.
3. **Let the Task Screener pick your day.** Study the top of the yield list, not what's comfortable.
4. **Grade yourself like a hostile examiner.** ROAST ME and MARK SCHEME exist to find marks you're losing now, in private, instead of in the hall.
5. **Match the mode to the job.** Fifteen specialists beat one generalist.
6. **Obey the command word.** Most lost marks are answers to a question that wasn't asked.
7. **Review before you add.** Due flashcards first, new study second — always.
8. **Trust the boxes, not the cram.** Spaced recall compounds; cramming evaporates.
9. **Watch Burn-Risk weekly.** It's the only honest read on whether you're winning the race against the clock.
10. **Protect the streak.** Small and logged beats heroic and forgotten.
11. **Fix the method when effort stops yielding.** The Effort-vs-Yield chart will tell you when grinding harder is the wrong answer.
12. **Be honest with the terminal.** It can only aim you as accurately as you report yourself.

The highest marks on the planet aren't taken in the exam hall. They're assembled, day by logged day, in this terminal — and then simply collected on the day.

Now close the manual, open SIEGE, and build today's mission. The loop is waiting.